

December 10th 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
 - three syntheses done with 0.02M sulfuric acid
 - first product not representative, not used further
 - second and third batch converted (see below)
 - four syntheses done with 0.10M sulfuric acid
 - very similar observations and outputs
 - products of these have been combined/homogenized
 - part of the combined lots has been converted
- Canadian Rockies Hemp arrived and cleared customs
 - several standard synthesis batches finished, seven of these (11.2 g) combined and homogenized
 - additional experiments done with higher concentrated sulfuric acid (1.5%, 2.0%, 5.0%)
 - one experiment performed under stirring, using cut up hemp fibers, product is finer and more homogeneous
 - one experiment performed with sulfuric acid pre-treatment of hemp (95 °C)
- Larger vessel reactivated, production ongoing (approx. 15 g output per batch)
- Typical moisture content 5% by KFT. Further drying at 100 °C under nitrogen stream decreases KFT result to 2%. Relevance unclear – being tested in step 2.
- Corrosion/cleaning test for large-scale equipment completed successfully, first formal meeting with pilot plant staff held
- Biochar found to be hygroscopic, drying at 40 °C results in 2% water, but content increases to approx. 5% upon standing under air

2. Biochar analysis

- Sample of combined biochar (milled, MB2946X_gem) analyzed by SEM and Raman for reference

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
 - biochar ground manually for first experiment (TB1131)
 - laboratory mill used for second experiment (TB1132)
 - products not yet characterized
 - no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min, TB1133/1134)
 - better handling but no other apparent difference
- one experiment performed at 5 °C/min (MB2946)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h: minimal yield (MB2947A)
 - 800 °C, 2 h: minimal yield (MB2947B)
 - 800 °C, 3 h: no output (MB2947C)
 - ➔ Inertion apparently insufficient and material likely incinerated
 - 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained (MB2947D)
 - 800 °C, 1 h, nitrogen inertion: 60 mg output (TB1135-1)
 - 800 °C, 1 h, increased nitrogen inertion: 70 mg output (TB1135-2)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 2 h, inertion equivalent to TB1135-2: 70 mg output (MB2952B)
 - 800 °C, 3 h, inertion equivalent to TB1135-2: 50 mg output (MB2952C)
 - ➔ smoldering observed upon contact of the cold material with air (see video)
 - 800 °C, 3 h, inertion equivalent to TB1135-2: 60 mg output (MB2952C2)
 - ➔ material wetted with water before/during removal from the apparatus

- Oxygen sensor shows complete inertion of apparatus within 3 minutes at selected nitrogen stream (equivalent to TB1135-2)

- 2 g of combined biochar milled with 3 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h, standard inertion: Quartz tube shattered, only 7 mg output (MB2955A)
 - 800 °C, 1 h, standard inertion (repeat of MB2955A): 90 mg output (MB2955A2)
 - 800 °C, 2 h, standard inertion: 80 mg output (TB1136)

- 2 g of combined biochar milled with 4 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h, standard inertion: 80 mg output (MB2962A)
 - 800 °C, 2 h, standard inertion: 90 mg output (MB2962B)

- 0.6 g of biochar (cut up, hydrolyzed under stirring) milled with 0.6 g of KOH and used for two batches on 0.2 – 0.3 g scale
 - 800 °C, 1 h, standard inertion: 70 mg output (MB2963A)
 - 800 °C, 2 h, standard inertion: 120 mg output (MB2963B)

- 2 g of combined biochar milled with 8 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h, standard inertion: 60 mg output (TB1137-1)

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- 800 °C, 2 h, standard inertion: 60 mg output (TB1137-2)
 - 800 °C, 1 h, standard inertion: 70 mg output (TB1137-3)

 - 2 g of biochar lot MB2959 milled with 12 g of KOH and used for several batches
 - 800 °C, 1 h, standard inertion: 24 mg output (MB2965A)
 - 800 °C, 2 h, standard inertion: 13 mg output (MB2965B)
 - 800 °C, 2 h, standard inertion (oven B): 7 mg output (MB2965B2)
 - 2 g of biochar lot MB2959 milled with 8 g of KOH and used for several batches
 - 800 °C, 1 h, standard inertion (oven B): 52 mg output (MB2966A)
 - 800 °C, 2 h, standard inertion: 46 mg output (MB2966B)

 - 1.2 g of biochar lot MB2964 milled with 4.8 g of KOH and used for several batches
 - 800 °C, 1 h, standard inertion (oven B): 57 mg output (MB2967A)
 - 800 °C, 2 h, standard inertion: 42 mg output (MB2967B)

 - 1.4 g of biochar lot MB2957 (from 1.5% H₂SO₄) milled with 2.1 g of KOH and used for several batches
 - 800 °C, 1 h, standard inertion (oven B): 63 mg output (MB2970A)
 - 800 °C, 2 h, standard inertion: 19 mg output (MB2970B)

 - 1.4 g of biochar lot MB2958 (from 2% H₂SO₄) milled with 2.1 g of KOH and used for several batches
 - 800 °C, 1 h, standard inertion (oven B): 54 mg output (MB2971A)
 - 800 °C, 2 h, standard inertion: 66 mg output (MB2971B)

 - 1.4 g of biochar lot MB2961 (from 5% H₂SO₄) milled with 2.1 g of KOH and used for several batches
 - 800 °C, 1 h, standard inertion (oven B): 88 mg output (MB2972A)
 - 800 °C, 2 h, standard inertion: 57 mg output (MB2972B)

 - 1.4 g of biochar lot MB2959 milled with 2.1 g of KOH and used for two identical batches (reproducibility experiment)
 - 800 °C, 1 h, standard inertion (oven B): 42 mg output (MB2973A)
 - 800 °C, 1 h, standard inertion: 90 mg output (MB2973B)

 - 1.5 g of biochar lot MB2959 milled with 2.3 g of KOH and used for two batches (milling time experiment)
 - milling time 5 min, 800 °C, 1 h, standard inertion (oven B): 78 mg output (MB2974A)
 - milling time 1 min, 800 °C, 1 h, standard inertion: 81 mg output (MB2974B)

 - 1.5 g of biochar lot MB2959 milled with 2.3 g of KOH and used for two batches (milling time experiment)
 - milling time 5 min, 800 °C, 2 h, standard inertion (oven B): 59 mg output (MB2975A)
 - milling time 1 min, 800 °C, 2 h, standard inertion: 80 mg output (MB2975B)

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- 2.0 g of biochar lot MB2959 milled with 3.0 g of KOH and used for two batches (pellet pressing experiment)
 - 0.6 g pellet, 800 °C, 1 h, standard inertion (oven B): 83 mg output (MB2976A)
 - 0.6 g pellet, 800 °C, 2 h, standard inertion: 108 mg output (MB2976B)

 - Biochar/KOH mixture from MB2976 used for new reproducibility experiment (analogous to MB2973)
 - 800 °C, 1 h, standard inertion (oven B): 83 mg output (MB2979A)
 - 800 °C, 1 h, standard inertion: 104 mg output (MB2979B)

 - 2.5 g of biochar lots MB2959/2956 milled with 3.8 g of KOH for 10/20 min and used for four batches (pellet pressing experiment)
 - 0.6 g pellet, 20 min milling time, 800 °C, 1 h, standard inertion (oven B): 78 mg output (MB2980A)
 - 0.6 g pellet, 10 min milling time, 800 °C, 1 h, standard inertion (oven A): 95 mg output (MB2980B)
 - 1.3 g pellet, 20 min milling time, 800 °C, 1 h, standard inertion (oven B): 163 mg output (MB2980C)
 - 1.3 g pellet, 10 min milling time, 800 °C, 1 h, standard inertion (oven A): 232 mg output (MB2980D)

 - 2.5 g of biochar lot MB2978 milled with 3.8 g of KOH for 10/20 min and used for ten batches
 - 800 °C, 1 h, 10 min milling time, standard inertion (oven A): 95 mg (KJo-0165A)
 - 800 °C, 1 h, 20 min milling time, inertion potentially nonstandard (oven B): 146 mg (KJo-0165B)
 - 800 °C, 1 h, 10 min milling time, 10 °C/min, standard inertion (oven A): 125 mg (KJo-0166A)
 - 800 °C, 1 h, 20 min milling time, 10 °C/min, standard inertion (oven B): 114 mg (KJo-0166B)
 - 800 °C, 4 h, 10 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 102 mg (KJo-0167A)
 - 800 °C, 4 h, 20 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 61 mg (KJo-0167B)
 - 900 °C, 2 h, 10 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 23 mg (KJo-0168A)
 - 900 °C, 1 h, 20 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 89 mg (KJo-0168B)
 - 800 °C, 1 h, 10 min milling time, 3 °C/min, 2.0 g pellet pressed, standard inertion (oven B): 232 mg (KJo-0169)
 - 800 °C, 1 h, 20 min milling time, 3 °C/min, 2.7 g pellet pressed, standard inertion (oven B): 619 mg (KJo-0170)

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- 5 g of biochar lot MB2978 milled with 7.5 g of KOH for 5 min and used for two batches
 - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven B): 642 mg (KJo-0171A)
 - 800 °C, 1 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 128 mg (KJo-0171B)
 - 2.5 g of biochar lot MB2978 milled with 2.5 g of KOH for 2.5 min and used for two batches
 - 800 °C, 1 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 243 mg (KJo-0172A)
 - 800 °C, 2 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 169 mg (KJo-0172B)
 - 3.0 g of biochar lot MB2978 milled with 4.5 g of KOH for 5 min and used for three batches
 - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven A), quartz tube shattered: 710 mg output (KJo-0175)
 - 800 °C, 1 h, 1.5 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 104 mg output (KJo-0176)
 - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven A): 908 mg output (KJo-0177)
 - Various combined lots of milled biochar/KOH (1:1.5, lots MB2955/2974/2975/2976) used for two production experiments
 - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, questionable inertion (oven A): 549 mg output (KJo-0178)
 - 800 °C, 1 h, 3 °C/min, 6 x 0.6 g pellets pressed, standard inertion (oven B): 441 mg output (MB2981)
 - Various combined lots of biochar (MB2946/2948/2949/2950/2951/2953/2956/MRa255) milled with KOH (1:1.5) – mixture heated itself moderately during milling
 - 800 °C, 1 h, 3 °C/min, 1.8 + 2.1 g pellets pressed, standard inertion (oven A): 587 mg output (TB1138)
 - 800 °C, 1 h, 3 °C/min, 1.8 + 1.7 g pellets pressed, standard inertion (oven B): 547 mg output (MRa320)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.8 g pellets pressed, standard inertion (oven B): 476 mg output (MB2984)
 - 800 °C, 1 h, 3 °C/min, 1.3 + 1.4 g pellets pressed, standard inertion (oven A): 395 mg output (TB1141)
 - 8.5 g biochar (KJo-0173) milled with KOH (1:1.5) – brown powder, quickly turned dark brown, somewhat chunky and hot (steaming) upon contact with air, product thus not perfectly homogeneous
 - 800 °C, 1 h, 3 °C/min, 2 x 1.8 g pellets pressed, standard inertion (oven A): 538 mg output (TB1139)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 464 mg output (TB1140A)

- 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 480 mg output (TB1140B)
- 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 421 mg output (MB2988)
- 5.0 g biochar (KJo-0173) milled, sonicated (1 h), milled with KOH (1:1.5) – brown powder, no noteworthy observations
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 464 mg output (MRa323)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 505 mg output (MRa326)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 459 mg output (MRa327)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.5 g pellets pressed, standard inertion (oven B): 528 mg output (MRa328)
- 1.1 g biochar (KJo-0173, extra dried at 100 °C) milled with KOH (1:1.5) – brown powder, no noteworthy observations
 - 800 °C, 1 h, 3 °C/min, 2 x 1.5 g pellets pressed, standard inertion (oven B): 434 mg output (MRa329) ➔ shinier, more voluminous product, color shifted to grey/silver
- 7.0 g biochar (MB2982) milled with KOH (1:1.5) – brown powder, turned dark brown, somewhat chunky and hot during/after bottling, manually homogenized
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 505 mg output (MB2990A)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 420 mg output (MB2990B)
 - biochar/KOH powder dried in N₂ stream (100 °C, 3 d), 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 524 mg output (TB1146)
- 7.0 g biochar (MB2985, extra dried at 100 °C) milled with KOH (1:1.5) – brown powder, turned dark brown, somewhat chunky and hot during/after bottling, manually homogenized
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 637 mg output (MRa331A) ➔ shinier, more voluminous product, color shifted to grey/silver
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 462 mg output (MRa331B) ➔ shinier, more voluminous product, color shifted to grey/silver
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 548 mg output (MB2997A)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 452 mg output (MB2997B)
- 7.0 g biochar (MB2985, extra dried at 100 °C) milled with KOH/NaOH (60/40, 1:1.5) – brown powder, stricter inertion after milling thus (?) no self-heating and darkening
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 212 mg output (MRa333A) ➔ shinier, more voluminous product, color shifted to grey/silver

- 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 258 mg output (MRa333B) → shinier, more voluminous product, color shifted to grey/silver
- 1:1 mixture of powders from MRa331 (biochar:KOH 1:1.5) and MRa333 (biochar:KOH:NaOH 1:0.9:0.6) → KOH:NaOH 0.8:0.2, used for two experiments
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 476 mg output (MRa334A) → shinier, more voluminous product, color shifted to grey/silver
 - 800 °C, 1 h, 3 °C/min, 2 x 1.1 g pellets pressed, standard inertion (oven B): 394 mg output (MRa334B) → shinier, more voluminous product, color shifted to grey/silver
- 7.0 g biochar (MB2982) milled with KOH/NaOH (60/40, 1:1.5) – brown powder, stricter inertion after milling thus (?) no self-heating and darkening
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 225 mg output (MB2993A) → shinier, more voluminous product, color shifted to grey/silver
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 189 mg output (MB2993B) → shinier, more voluminous product, color shifted to grey/silver
- 7.0 g biochar (MB2983, extra dried at 100 °C) milled with KOH (1:1.5) – brown powder, part of the mixture started smoldering during bottling and was lost, the bottled material was not affected and remained fine brown powder
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 536 mg output (MB2995A)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 457 mg output (MB2995B)
 - biochar/KOH powder dried in N₂ stream (100 °C, 3 d), 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 355 mg output (TB1147)
- 2.8 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (15 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven A): 613 mg output (TB1144A)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 355 mg output (TB1144B)
- 3.4 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (25 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 676 mg output (MB2999A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 400 mg output (MB2999B)
- 3.4 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (35 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 493 mg output (MB3001A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 478 mg output (MB3001B)

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- 5.5 g 2% wet biochar (MB2986) milled with KOH (1:1.5) in ball mill (35 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 574 mg output (MB3003A) → powder used fresh
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 435 mg output (MB3003B) → powder used fresh
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 592 mg output (MB3004A) → powder used after standing for a day
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 406 mg output (MB3004B) → powder used after standing for a day

 - 4.6 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (15 Hz, 10 min) – brown powder, appeared homogeneous, limited darkening noticed during bottling under nitrogen
 - 800 °C, 1 h, 3 °C/min, 1.7/1.8 g pellets pressed, standard inertion (oven A): 531 mg output (TB1151A)
 - 800 °C, 1 h, 3 °C/min, 2.7 g powder, standard inertion (oven B): 450 mg output (TB1151B)

 - 4.3 g biochar (KJo-0174/MB2982/MB2989) milled with KOH (1:1.5) in ball mill (15 Hz, 30 min) – brown powder, appeared homogeneous, limited darkening noticed during bottling under nitrogen
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 501 mg output (MB3007A) → second pellet disintegrated, only parts used for reaction (attempted twice)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 433 mg output (MB3007B)

 - 3.0 g biochar (MB2992, reaction temperature 160 °C) milled with KOH (1:1.5) – brown powder, bottled without change in appearance
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 405 mg output (MB3009)

 - 4.7 g biochar (MB2983, 2% water) milled in ball mill (15 Hz, 10 min), then milled with KOH (1:1.5) for 1 min – brown powder, appeared homogeneous
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 440 mg output (MB3012B)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 400 mg output (MB3013B)

 - 4.2 g biochar (MB2983, 2% water) milled in ball mill (35 Hz, 10 min), then milled with KOH (1:1.5) for 1 min – fine brown powder, appeared homogeneous
 - 800 °C, 1 h, 3 °C/min, 3.2 g powder, standard inertion (oven A): 639 mg output (MB3012A)
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 577 mg output (MB3013A)

 - 4.2 g biochar (MB2986, 2% water) milled in ball mill (15 Hz, 30 min), then milled with KOH (1:1.5) for 1 min – fine brown powder, appeared homogeneous
 - 800 °C, 1 h, 3 °C/min, 2 x 1.5 g pellets pressed, standard inertion (oven A): 391 mg output (MB3015A)
 - 800 °C, 1 h, 3 °C/min, 2.5 g powder, standard inertion (oven B): 355 mg output (MB3015B)

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- 6.0 g biochar (MB2996) milled with KOH (1:1.5) in ball mill (15 Hz, 120 min) – fine brown powder, some darkening during bottling
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 488 mg output (MRa340A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 441 mg output (MRa340B)

 - 4.9 g biochar (MB2986, 2% water) milled with KOH/NaOH (1:1.2:0.3) in ball mill (15 Hz, 10 min) – fine brown powder, unmilled NaOH pellets discovered in powder after having started part A
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 690 mg output (TB1155A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder (leftover pellets removed), standard inertion (oven B): 451 mg output (TB1155B)

 - 4.2 g biochar (MB3002, 2% water) milled with KOH/NaOH (1:1.2:0.3) in ball mill (35 Hz, 10 min) – fine brown powder, appeared homogeneous
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 527 mg output (MB3017A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 332 mg output (MB3017B)

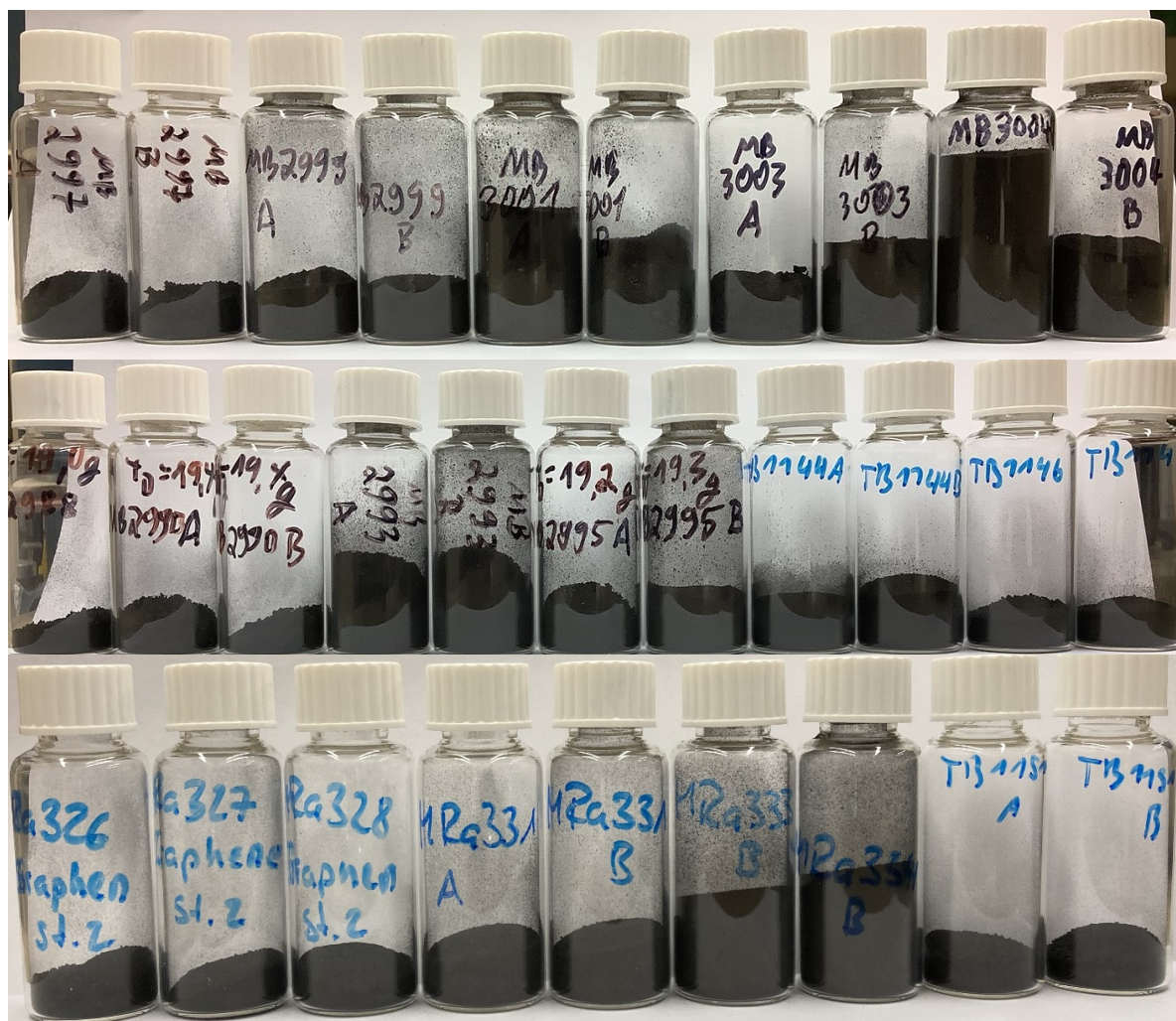
 - 5.2 g biochar (MB3002, 2% water) milled in ball mill (15 Hz, 10 min), then milled with KOH (1:1.5) for 1 min – fine brown powder, appeared homogeneous
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 469 mg output (MB3019A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 392 mg output (MB3019B)

 - 3.0 g biochar (MB3002, 2% water) milled 2 min with KOH (1:1.5) – brown powder, bottled without change in appearance
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 522 mg output (MB3021A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 433 mg output (MB3021B)

 - 5.0 g biochar (MB2989) milled with KOH/NaOH (1:1.2:0.3) in ball mill (35 Hz, 10 min) – fine brown powder, appeared homogeneous
 - 800 °C, 1 h, 3 °C/min, 2 x 1.35 g pellets pressed, standard inertion (oven A): 279 mg output (MB3023A)
 - 800 °C, 1 h, 3 °C/min, 2.0 g powder, standard inertion (oven B): 247 mg output (MB3023B)

 - 5.0 g biochar (MB3002, 2% water) milled with KOH (1:1.5) in ball mill (35 Hz, 10 min) – brown powder, bottled without change in appearance
 - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 619 mg output (MRa352A)
 - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 526 mg output (MRa352B)

- biochar (MB2991) milled with KOH and NaOH in different amounts (in total 1:1.5) in ball mill (35 Hz, 10 min) – brown powders, bottled without change in appearance
 - 800 °C, 1 h, 3 °C/min, 0% NaOH, 2 x 1.6 g pellets pressed, standard inertion (oven A): 528 mg output (MRa353A)
 - 800 °C, 1 h, 3 °C/min, 10% NaOH, 2 x 1.3 g pellets pressed, standard inertion (oven B): 442 mg output (MRa353B)
 - 800 °C, 1 h, 3 °C/min, 20% NaOH, 2 x 1.3 g pellets pressed, standard inertion (oven A): 402 mg output (TB1158A)
 - 800 °C, 1 h, 3 °C/min, 5% NaOH, 2 x 1.2 g pellets pressed, standard inertion (oven B): 387 mg output (TB1158B)
 - 800 °C, 1 h, 3 °C/min, 20% NaOH, 2 x 1.3 g pellets pressed, standard inertion (oven A): 460 mg output (TB1160A)
 - 800 °C, 1 h, 3 °C/min, 0% NaOH, 2 x 1.3 g pellets pressed, standard inertion (oven B): 431 mg output (TB1160B)



- MB2976 A/B: approx. 100 mg
- MB2980 A: approx. 50 mg
- MB2980 C/D: approx. 300 mg
- KJo-0166 B: approx. 100 mg
- KJo-0167 B: approx. 50 mg
- KJo-0168 B: approx. 70 mg
- KJo-0171 A: approx. 600 mg
- KJo-0171 B: approx. 100 mg
- KJo-0172 A: approx. 220 mg
- KJo-0175: approx. 600 mg
- KJo-0176: approx. 80 mg
- KJo-0177: approx. 800 mg
- KJo-0178: approx. 500 mg
- MB2981: approx. 400 mg

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- TB1138: approx. 500 mg
 - MRa320: approx. 500 mg
 - MB2984: approx. 400 mg
 - TB1139: approx. 500 mg
 - TB1140A: approx. 400 mg
 - TB1140B: approx. 400 mg
 - MRa323: approx. 400 mg
 - TB1141: approx. 300 mg

4. Graphene analysis

- SEM (and partially Raman) results available for
 - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B, MB2974A/B, MB2975A/B, MB2976A/B, MB2979A/B, MB2980A/B/C/D, KJo-0165A/B, KJo-0166A/B, KJo-0167A/B, KJo-0168A/B, KJo-0169, KJo-0170, KJo-0171A/B, KJo-0172A/B, KJo-0175, KJo-0176, KJo-0177, KJo-0178, KJo-0171A GR, MB2981A, TB1138, MB2984, MRa320, TB1139, TB1140A, TB1140B, MB2988, MRa323, TB1141, MRa326, MRa327, MRa328, MB2990A, MB2990B, MRa329, MRa331A, MRa331B, MRa333A, MRa333B, MRa334A, MRa334B, MRa2993A, MRa2993B, MB2995A, MB2995B, MB2997A, MB2997B, TB1144A, MB1144B, MB2999A, MB2999B, TB1146, TB1147, MB3001A, MB3001B, MB3003A, MB3003B, MB3004A, MB3004B, TB1151A, TB1151B, MB3007A, MB3007B, MB3009, MB3013A, MB3013B, MB3015A, MB3015B, MRa340A, MRa340B, MB3017A, MB3017B, MB3019A, MB3019B, MB3021A, MB3021B, MB3023A, MB3023B, MRa352A, MRa352B, MRa353A, MRa353B, TB1158A, TB1158B, TB1142 (step 2 products)
 - 900561-500MG, 2 lots (Sigma-Aldrich reference)
 - 924458-500MG (Sigma-Aldrich reference “Single-layer graphene sheets for battery, Bio-sourced”)
 - Graphene oxide
- TEM results available for
 - MB2955A2 (step 2 product)
 - MB2976A (step 2 product)
 - TB1137-1 (step 2 product)

- BET results

Sample	Multipoint BET Area [m ² /g]	Langmuir Surface Area [m ² /g]
MB2976A	1.88 x 10 ³	2.55 x 10 ³
MRa329	1.76 x 10 ³	2.39 x 10 ³
MRa333A	2.09 x 10 ³	5.74 x 10 ³
MRa334A	1.86 x 10 ³	2.50 x 10 ³

5. Currently open questions

- Patent: new process/patent
 - So far mainly focusing on pellet (weak)
 - Needs to brainstorm/think about other improvements/new ideas
- Characteristics of the Graphene
 - BET results
 - RAMAN results

- Electrical conductivity
- Volume fraction of mesoporosity
- Starting discussion on next size commercial production
 - Group call pending
- Is partial replacement of KOH by NaOH or LiOH beneficial?
- Does the shiny, voluminous material (as obtained from MRa329) have desirable properties?
 - Which parameters lead to the formation of this material?
- Does the use of a ball mill lead to different results compared to the previously used “coffee grinder” type mill?
 - What effect do different milling parameters have on the properties of the product?
- How to mill and press larger quantities of biochar during scaleup?

6. Questions Answered

- Are we able to reproduce Dr Li's extraction
 - By SEM: yes (based on SEM at Dr Li's SEM)
 - By TEM: result needed to confirm
- Are we able to produce 3-5g of Graphene
 - Done (TB1142, 7.7 g)
- What are the next steps to Scale-up
 - Step 1: Decision on reaction vessel, corrosion/cleaning test
 - Step 2: Reproduction of previous results with rotating oven
- Rotating oven
 - Arrived, being set up

Experiment overview – Step 1

Experiment	Reactor	Raw material			Acid			T	t	pressure		Wash		Output		
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 10.4 bar	742 g	Water	2.6 g		
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	-> 9.5 bar	647 g	Water	1.2 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	-> 9.5 bar	500 g	Water	1.3 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	-> 10.6 bar	589 g	Water	0.8 g	Combined and homogenized as lot no MRa231	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	-> 10.5 bar	500 g	Water	0.9 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 11.0 bar	500 g	Water	0.9 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	-> 10.5 bar	500 g	Water	0.8 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	-> 10.9 bar	569 g	Water	1.6 g	Combined and homogenized as lot no MB2946X	
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	-> 13.2 bar	577 g	Water	1.4 g		
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 10.6 bar	555 g	Water	1.5 g		
Comment	very low amount of foreign material observed and removed during filtration															
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	-> 13.5 bar	578 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	-> 12.7 bar	611 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	-> 9.7 bar	593 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	-> 9.8 bar	6 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	-> 12.6 bar	475 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid			T	t	pressure		Wash		Output	Drying	KFT
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M Sulfuric acid	180 °C	23.5 h	8.4 bar	-> 10.2 bar	538 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M Sulfuric acid	180 °C	23 h		-> 11.1 bar	534 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M Sulfuric acid	180 °C	24 h	9.1 bar	-> 11.2 bar	575 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M Sulfuric acid	180 °C	23.5 h	8.1 bar	-> 12.8 bar	488 g	Water	1.7 g		
Comment	some foreign material observed and removed during filtration														
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M Sulfuric acid	180 °C	23.5 h	9.1 bar	-> 10.4 bar	558 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M Sulfuric acid	180 °C	24 h	8.6 bar	-> 12.5 bar	548 g	Water	1.9 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M Sulfuric acid	180 °C	23.5 h	8.8 bar	-> 8.5 bar	589 g	Water	1.5 g		
Comment															
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M Sulfuric acid	180 °C	24 h	8.9 bar	-> 8.7 bar	562 g	Water	1.8 g		
Comment															
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M Sulfuric acid	180 °C	25 h			3957 g	Water	14.9 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M Sulfuric acid	180 °C	24 h	8.8 bar	-> 10.5 bar	300 g	Water	0.6 g		
Comment	Reaction performed with stirring														
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M Sulfuric acid	180 °C	24 h	10.1 bar	-> 16.1 bar	563 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M Sulfuric acid	180 °C	24 h	8.6 bar	-> 8.8 bar	310 g + 603 g	Water	1.3 g		
Comment	Hemp boiled with 1% H2SO4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid														
MB2978	AV5	76.3 g	Canadian Rockies Hemp	822.0 g	1.00%	0.10 M Sulfuric acid	180 °C	22 h			4112 g	Water	15.4 g		
Comment															
KJo-0173	AV5	77.1 g	Canadian Rockies Hemp	942.1 g	1.00%	0.10 M Sulfuric acid	185 °C	24 h			4283 g	Water	16.5 g	40 °C	4.9%
Comment														100 °C	2.3%
KJo-0174	AV5	77.1 g	Canadian Rockies Hemp	945.0 g	1.00%	0.10 M Sulfuric acid	190 °C	24 h			4384 g	Water	15.8 g	40 °C	3.7%
Comment															
MB2982	AV5	77.1 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M Sulfuric acid	190 °C	24 h			4258 g	Water	14.9 g	40 °C	5.3%
Comment															
MB2983	AV5	80.5 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M Sulfuric acid	190 °C	24 h			4232 g	Water	16.6 g	40 °C	5.5%
Comment														100 °C	1.7%

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB2985	AV5	79.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	200 °C	24 h		4241 g Water	15.4 g	40 °C	3.7%
Comment									100 °C	1.9%
MB2986	AV5	79.9 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4128 g Water	16.0 g	60 °C	2.7%
Comment									100 °C	1.7%
MB2989	AV5	80.2 g Canadian Rockies Hemp	999.3 g 1.00% 0.10 M Sulfuric acid	180 °C	23 h		4314 g Water	18.2 g	40 °C	5.9%
Comment										
MB2991	AV5	81.3 g Canadian Rockies Hemp	999.3 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4223 g Water	19.1 g	40 °C	5.7%
Comment										
MB2992	AV5	81.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	160 °C	23.5 h		4262 g Water	19.5 g	40 °C	6.2%
Comment										
MB2994	AV5	80.8 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4159 g Water	20.2 g	40 °C	5.2%
Comment										
MB2996	AV5	82.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4210 g Water	20.0 g	40 °C	4.6%
Comment										
MB2998	AV5	84.0 g Canadian Rockies Hemp	999.6 g 1.00% 0.10 M Sulfuric acid	180 °C	22 h		4263 g Water	19.8 g	40 °C	5.0%
Comment	Included nickel and steel material samples for corrosion/cleaning tests									
MB3000	AV5	82.5 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4431 g Water	19.8 g	40 °C	5.0%
Comment										
MB3002	AV5	80.0 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	190 °C	25 h		4245 g Water	18.5 g	40 °C	2.0%
Comment										
MB3005	AV5	81.4 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	200 °C	24 h		4280 g Water	19.0 g	40 °C	2.0%
Comment										
MB3006	AV5	80.8 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	190 °C	23 h		4182 g Water	17.4 g	40 °C	2.1%
Comment										
MB3008	AV5	82.2 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4238 g Water	18.0 g	40 °C	2.6%
Comment										
MB3010	AV5	81.3 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4235 g Water	18.9 g	40 °C	2.0%
Comment										
MB3011	AV5	82.4 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	160 °C	24 h		4244 g Water	23.5 g	40 °C	2.2%
Comment										
MB3014	AV5	82.8 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		1460 g Water	23.6 g	40 °C	1.3%
Comment										
MRa341	AV5	82.3 g Canadian Rockies Hemp	1000.4 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		1232 g Water	19.2 g	40 °C	
Comment										

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB3016	AV5	81.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	21 h		4232 g Water	19.7 g	40 °C	2.1%
Comment										
MB3018	AV5	81.2 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	195 °C	22 h		4266 g Water	19.0 g	40 °C	2.1%
Comment										
MB3020	AV5	81.1 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4278 g Water	18.9 g	40 °C	1.7%
Comment										
MB3022	AV5	81.8 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4262 g Water	19.0 g	40 °C	2.9%
Comment										
MB3024	AV5	80.7 g Canadian Rockies Hemp	999.6 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4256 g Water	18.4 g	40 °C	
Comment										
MRa349	AV5	80.8 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4254 g Water		40 °C	
Comment										
MRa354	AV5	80.2 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4414 g Water		40 °C	
Comment										
MRa355	AV5	80.2 g Canadian Rockies Hemp	999.5 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4262 g Water		40 °C	
Comment										
MRa358	AV5	80.8 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		Water		40 °C	
Comment										

Comments

- Red text for T and t shows experiments where the temperature tracks have not yet been evaluated (or are not available)

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
TB1131	A	1.2 g	MB2933	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										
TB1132	A	1.3 g	MB2935	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder										
TB1133	A	1.0 g	MRa231	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
TB1134	A	1.0 g	MRa231	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
MB2946	A	1.0 g	MB2946X	1.0 g KOH (90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947A	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947B	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947C	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded		
Comment		ground biochar: brown powder; see temperature trend										
MB2947D	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	38 mg
Comment		ground biochar: brown powder; see temperature trend										
TB1135-1	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.06 g
Comment		ground biochar: brown powder; see temperature trend										
TB1135-2	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.07 g
Comment		ground biochar: brown powder; see temperature trend										
MB2952B	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.07 g
Comment		ground biochar: brown powder; see temperature trend										
MB2952C	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.05 g
Comment		ground biochar: brown powder; see temperature trend										

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2952C2	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2955A	A	0.25 g	MB2946X	0.38 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	7 mg
Comment		Quartz tube shattered during experiment								(+ water)	(atm. Pressure)	
MB2955A2	A	0.25 g	MB2946X	0.38 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1136	A	0.26 g	MB2946X	0.39 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	14 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2962A	A	0.25 g	MB2946X	0.50 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2962B	A	0.25 g	MB2946X	0.50 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2963A	A	0.22 g	MB2960	0.22 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2963B	A	0.29 g	MB2960	0.29 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-1	A	0.26 g	MB2946X	1.06 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-2	A	0.28 g	MB2946X	1.13 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	23 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-3	A	0.26 g	MB2946X	1.06 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2965A	A	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.02 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	
MB2965B	A	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	
MB2965B2	B	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2966A	B	0.26 g	MB2959	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2966B	A	0.26 g	MB2959	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2967A	B	0.26 g	MB2964	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2967B	A	0.26 g	MB2964	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2970A	B	0.24 g	MB2957	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2970B	A	0.24 g	MB2957	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.02 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2971A	B	0.24 g	MB2958	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2971B	A	0.24 g	MB2958	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2972A	B	0.24 g	MB2961	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2972B	A	0.24 g	MB2961	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2973A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2973B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2974A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2974B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2975A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2975B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2976A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend								(+ water)	(atm. Pressure)	
MB2976B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.11 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend								(+ water)	(atm. Pressure)	
MB2979A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2979B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.10 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2980A	B	0.24 g	MB2959/2956	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980B	A	0.24 g	MB2959/2956	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980C	B	0.52 g	MB2959/2956	0.78 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	21 g 10% HCl	100 °C N2 stream	0.16 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980D	A	0.52 g	MB2959/2956	0.78 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	21 g 10% HCl	100 °C N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
KJo-0165A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0165B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.15 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0166A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	10 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.13 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0166B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	10 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.11 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
KJo-0167A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	4 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g
		ground biochar (brown powder) compacted to pellet										
KJo-0167B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	4 h	11 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.06 g
		ground biochar (brown powder) compacted to pellet										
KJo-0168A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	900 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.02 g
		ground biochar (brown powder) compacted to pellet										
KJo-0168B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	900 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.09 g
		ground biochar (brown powder) compacted to pellet										
KJo-0169	B	0.80 g	MB2978	1.20 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	30 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.23 g
		ground biochar (brown powder) compacted to pellet										
KJo-0170	B	1.07 g	MB2978	1.61 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	38 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.62 g
		ground biochar (brown powder) compacted to pellet										
KJo-0171A	B	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g
		ground biochar (brown powder) compacted to two pellets of equal size										
KJo-0171B	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.13 g
		ground biochar (brown powder) compacted to pellet										
KJo-0172A	B	0.50 g	MB2978	0.50 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	11 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.24 g
		ground biochar (brown powder) compacted to pellet										
KJo-0172B	A	0.50 g	MB2978	0.50 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.17 g
		ground biochar (brown powder) compacted to pellet										
KJo-0175	A	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.71 g
		ground biochar (brown powder) compacted to two pellets of equal size, quartz tube shattered										
KJo-0176	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (5 min)	N2	1.5 °C/min	800 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g
		ground biochar (brown powder) compacted to pellet										
KJo-0177	A	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.91 g
		ground biochar (brown powder) compacted to two pellets of equal size										
KJo-0178	A	1.60 g	MB2955/74/75/76	2.40 g KOH (90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										
MB2981	B	1.44 g	MB2955/74/75/76	2.16 g KOH (90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g
		ground biochar (brown powder) compacted to six pellets of equal size										

Experiment	Oven	Qty	Lot	Base	grinding	homogen?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
TB1138	A	1.54 g	various	2.31 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	59 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.59 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa320	B	1.43 g	various	2.14 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2984	B	1.40 g	various	2.10 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 34%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1139	A	1.39 g	KJo-0173	2.08 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.54 g 39%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1140A	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	49 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 36%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1140B	B	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa323	A	1.06 g	KJo-0173	1.60 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 41%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
TB1141	A	1.08 g	various	1.62 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	30 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 37%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa326	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.51 g 39%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MRa327	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MB2988	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa328	B	1.20 g	KJo-0173	1.80 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 44%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MRa329	A	1.04 g	KJo-0173 (extra dried)	1.56 g KOH (90%)	mill (1 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2990A	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.50 g 39%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2990B	B	1.04 g	MB2982	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogen?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
MRa331A	B	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g 50%		Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa331B	A	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%		Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa333A	A	1.28 g	MB2985 (extra dried)	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.21 g 17%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa333B	B	1.04 g	MB2985 (extra dried)	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.26 g 25%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa334A	A	1.04 g	MB2985 (extra dried)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa334B	B	0.88 g	MB2985 (extra dried)	1.06 g KOH (90%) 0.26 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	33 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 45%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2993A	A	1.28 g	MB2982	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.22 g 18%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2993B	B	1.04 g	MB2982	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.19 g 18%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2995A	A	1.28 g	MB2983 (extra dried)	1.92 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.54 g 42%	3.9 ml	Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2995B	B	1.04 g	MB2983 (extra dried)	1.56 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%	5.6 ml	1/2 mix	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1144A	A	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	38 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.61 g 59%	5.0 ml	1 + fibres	black/grey somewhat shiny
		ground biochar (brown powder) NOT compacted														
TB1144B	B	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	54 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 28%	6.0 ml	2 (?)	black/grey shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2997A	A	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g 43%	3.6 ml	1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2997B	B	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	40 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 43%	2.9 ml	1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2999A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.68 g 53%	4.6 ml	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogen?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
MB2999B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 38%	4.6 ml	1 + fibres	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted													
TB1146	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	51 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.52 g 41%	4.0 ml	1	black/grey somewhat shiny
			ground biochar/KOH mix (brown powder, MB2990) dried at 100 °C, then compacted to two pellets of equal size													
TB1147	B	1.08 g	MB2983 (extra dried)	1.62 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	37 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 33%	5.0 ml	Mostly 1	black/grey somewhat shiny
			ground biochar/KOH mix (brown powder, MB2995) dried at 100 °C, then compacted to two pellets of equal size													
MB3001A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.49 g 38%	10.6 ml	Mostly 2	shiny black/grey very voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MB3001B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%	8.2 ml	Mostly 2	shiny black/grey very voluminous
			ground biochar (brown powder) NOT compacted													
MB3003A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.57 g 45%	5.7 ml	Mostly 1	black/grey somewhat shiny
			freshly ground biochar (brown powder) compacted to two pellets of equal size													
MB3003B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 42%	9.0 ml	Mostly 1/2 mix	shiny black/grey voluminous
			freshly ground biochar (brown powder) NOT compacted													
MB3004A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.59 g 46%	17.4 ml	Mostly 2	shiny black/grey very voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MB3004B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 39%	8.4 ml	Mostly 1/2 mix	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted													
TB1151A	A	1.40 g	KJo-0174	2.10 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	51 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 38%	2.5 ml	1	black/grey brittle
			ground biochar (brown powder) compacted to two pellets of equal size													
TB1151B	B	1.08 g	KJo-0174	1.62 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 42%	2.5 ml	1	black/grey brittle
			ground biochar (brown powder) NOT compacted													
MB3007A	A	1.20 g	KJo-0174	1.80 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.50 g 42%	3.0 ml	1	black/grey barely shiny
			ground biochar (brown powder) compacted to two pellets of equal size													
MB3007B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%	2.9 ml	1	black/grey barely shiny
			ground biochar (brown powder) NOT compacted													
MB3009	A	1.28 g	MB2992	1.92 g KOH (90%)	mill (1 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 32%	3.0 ml	1	black/grey brittle
			ground biochar (brown powder) compacted to two pellets of equal size													
MB3012A	A	1.28 g	MB2983 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g 50%	10.2 ml	1/2 mix	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted													

Experiment	Oven	Qty	Lot	Base	grinding	homogen?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance			
MB3012B	B	1.04 g	MB2983 (2% water)	1.56 g KOH (90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 42%	10.8 ml	Mostly 1/2 mix	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted																
MB3013A	A	1.28 g	MB2983 (2% water)	1.92 g KOH (90%)				ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.58 g 45%	4.3 ml	Mostly 1/2 mix	shiny black/grey
			ground biochar (brown powder) compacted to two pellets of equal size																
MB3013B	B	1.04 g	MB2983 (2% water)	1.56 g KOH (90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 38%	12.8 ml	Mostly 2	shiny black/grey very voluminous
			ground biochar (brown powder) compacted to two pellets of equal size																
MB3015A	A	1.20 g	MB2986 (2% water)	1.80 g KOH (90%)				ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 33%	8.0 ml	Mostly 2	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size																
MB3015B	B	1.00 g	MB2986 (2% water)	1.50 g KOH (90%)				ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 36%	8.0 ml	Mostly 2	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted																
MRa340A	A	1.28 g	MB2996	1.92 g KOH (90%)				ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.49 g 38%	4.0 ml	1	black/grey barely shiny
			ground biochar (brown powder) compacted to two pellets of equal size																
MRa340B	B	1.04 g	MB2996	1.56 g KOH (90%)				ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 42%	3.5 ml	1	black/grey barely shiny
			ground biochar (brown powder) NOT compacted																
TB1155A	A	1.28 g	MB2986 (2% water)	1.54 g KOH (90%)	0.38 g NaOH (98%)			ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.69 g 54%	4.0 ml		black/grey somewhat shiny
			NaOH not fully ground!, ground biochar (brown powder) compacted to two pellets of equal size																
TB1155B	B	1.04 g	MB2986 (2% water)	1.25 g KOH (90%)	0.31 g NaOH (98%)			ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 43%	4.0 ml		black/grey somewhat shiny
			NaOH not fully ground!, ground biochar (brown powder) NOT compacted																
MB3017A	A	1.28 g	MB3002 (2% water)	1.54 g KOH (90%)	0.38 g NaOH (98%)			ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 41%	19.1 ml	2	shiny black/grey very voluminous
			ground biochar (brown powder) compacted to two pellets of equal size																
MB3017B	B	1.04 g	MB3002 (2% water)	1.25 g KOH (90%)	0.31 g NaOH (98%)			ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.33 g 32%	12.2 ml	2	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted																
MB3019A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.47 g 37%	15.9 ml	2	shiny black/grey very voluminous
			ground biochar (brown powder) compacted to two pellets of equal size																
MB3019B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 38%	9.3 ml	Mostly 2	shiny black/grey voluminous
			ground biochar (brown powder) NOT compacted																
MB3021A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)				mill (2 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.52 g 41%	7.3 ml	Mostly 1	shiny black/grey
			ground biochar (brown powder) compacted to two pellets of equal size																
MB3021B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)				mill (2 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%	5.3 ml	Mostly 1	shiny black/grey
			ground biochar (brown powder) NOT compacted																

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
MB3023A	A	1.08 g	MB2989	1.30 g KOH (90%) 0.32 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.28 g 26%	16.5 ml	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3023B	B	0.80 g	MB2989	0.96 g KOH (90%) 0.24 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.25 g 31%	13.5 ml	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) NOT compacted														
MRa352A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.62 g 48%	2.9 ml	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa352B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 51%	2.8 ml	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted														
MB353A	A	1.28 g	MB2991	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 41%	2.0 ml	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa353B	B	1.04 g	MB2991	1.39 g KOH (90%) 0.17 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 43%	3.5 ml	Mostly 1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1158A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 39%	17.0 ml	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1158B	B	0.96 g	MB2991	1.37 g KOH (90%) 0.08 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 40%	3.0 ml	Mostly 1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1160A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%	17.0 ml		shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1160B	B	1.04 g	MB2991	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	41 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 41%	2.0 ml		black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														

Comments

- Yellow/red text for the lot of biochar show cases where the quality of the milled material was somewhat/significantly impaired by contact with air and/or moisture (leading to heating or colour change of the material, stickiness, clumping or similar changed in appearance)
- The information on homogeneity of the milled material is based on optical and acoustic evaluation